

DINESH CHAUDHARI

Full Stack Developer

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SUMMARY

- Full Stack Developer who owns products end to end — from Firebase-backed Node.js APIs and third-party integrations through to React 19 web apps shipped to Android via Capacitor.
- Sole developer of a production logistics platform serving 350+ active users, spanning courier aggregation, payments, WhatsApp/push messaging and a custom over-the-air mobile update pipeline.
- Comfortable across the stack — REST API design, payment and refund ledgers, real-time systems (WebRTC, Socket.IO, CRDTs) and practical AI integration.

EXPERIENCE

Full Stack Developer

Innover Labs

Jul 2025 - Present Pune, Maharashtra

Rabbitship — production multi-courier shipping & logistics platform (350+ active users)

- Sole developer of the entire platform: a React 19 + TypeScript customer app shipped to Android via Capacitor (26 screens), a 31-screen super-admin operations console, and a Node.js / Express 5 + Firebase backend (~30K LOC across 124 modules and 20 REST route groups).
- Integrated three courier and delivery providers (Icarr, Shipmozo, Porter 2W same-city instant delivery) behind a single quoting and booking abstraction, with provider-specific pricing markup, pincode serviceability checks and Google Maps geocoding for on-demand pickups.
- Built the full payment stack on Cashfree — checkout, wallet, refund ledgers, coupon engine, referral cashback and partner commission payouts — with signature-verified, idempotent webhook handling for status, NDR and weight-dispute events.
- Designed a custom OTA update system that ships new web bundles straight to installed Android devices without a Play Store release (currently v1.0.16), gated by native versionCode and a minimum-supported-version policy so breaking releases force a native upgrade.
- Automated operations with a Firestore-backed node-cron scheduler running 6 recurring jobs — daily pickup PDF reports, refund-pending alerts, support-SLA escalation and WhatsApp template approval polling — each re-schedulable from the admin UI and manually triggerable without disturbing its next tick.
- Implemented multi-channel customer messaging: WhatsApp Business API driven by a versioned template registry, Web Push (VAPID), SMS and transactional email, plus a support ticket system with threaded chat and media attachments.
- Generated shipping labels and daily reports as server-side PDFs (pdf-lib + fontkit) and moved all user media to Backblaze B2 over the S3 API behind time-limited presigned URLs.
- Added AI-assisted address capture — users photograph a parcel label, business card or handwritten note and an OpenAI vision call returns a validated, structured Indian address (flat / area / city / PIN / state), reducing a multi-field form to a single tap.

Technology Stack : React 19, TypeScript, Tailwind CSS 4, shadcn/ui, TanStack Query, Capacitor, Node.js, Express 5, Firebase (Firestore, Auth, Admin SDK), Cashfree, AWS S3 SDK, node-cron, pdf-lib, OpenAI API

CORE STRENGTHS

- End-to-end product ownership
- REST API & schema design
- Payments, refunds & webhooks
- Real-time & offline sync
- Mobile delivery (Capacitor, OTA)
- Applied AI integration

DOMAIN EXPERTISE

- Logistics & Shipping
- Payments & FinTech
- Real-time Collaboration
- IoT & Deep Learning
- Messaging & Notifications

AREA OF EXPERTISE

- Web Applications
- Android Applications
- REST APIs & Webhooks
- Single Page Applications
- Embedded Firmware (ESP32)

PLATFORMS & CLOUD

Firebase	Advanced
Node.js / Express	Advanced
React / TypeScript	Advanced
AWS S3 / Backblaze B2	Proficient
Render / Vercel	Proficient

EDUCATION

MCA

Manipal University Jaipur

2025 - Present

B.Sc. Computer Science

Modern College, Shivajinagar, Pune

2022 - 2025

GPA 9.8 / 10 — 87%

LANGUAGES

English | Hindi | Marathi
Japanese (N5) | Marwadi

GhostDesk

Disposable Collaborative Workspace — zero persistence, no database

[ghostdesk.probuilder.in](#) | [github.com/Dinesh99673/GhostDesk](#)

14th place (Top 20) — OpenAI Codex Hackathon, OpenAI × NamasteDev

- Built a zero-persistence collaborative workspace — video calls, chat, notes, a code editor, a whiteboard and file transfer — where every room lives only in server RAM and is permanently destroyed 30 seconds after the last participant leaves. No signup, no database, no content logs.
- Implemented full-mesh WebRTC with perfect negotiation for up to 6 participants, plus TURN relay fallback so calls connect across symmetric NAT and mobile carrier networks.
- Powered conflict-free simultaneous editing with Yjs CRDTs shared by both the notes panel and a CodeMirror 6 editor, with in-browser PDF export (jsPDF) and one-click code execution.
- Streamed files peer-to-peer over WebRTC data channels in 64 KB chunks with backpressure handling — file bytes never reach the server.
- Added a shared push-to-talk AI agent (OpenAI speech-to-text → chat model with tools → text-to-speech) that can append to the room's live notes or propose code changes for human approval.
- Fit the whole system inside a 512 MB instance using per-room memory caps with automatic trimming; deployed on Render as a single Express service, verified by an end-to-end Socket.IO contract smoke test.

Technology Stack : React 19, TypeScript, Node.js, Express, Socket.IO, WebRTC, Yjs, CodeMirror 6, Excalidraw, Tailwind CSS 4, OpenAI API

Gesture-AI

Air-Writing Recognition System — IoT + Deep Learning

[Project demo](#) | [github.com/Dinesh99673/gesture-ai](#)

- Built an end-to-end system that lets a user write letters in mid-air: a glove-mounted ESP32-C3 + MPU-6050 IMU streams motion over serial, an LSTM classifies the character, and a desktop GUI assembles words and sentences and speaks them aloud.
- Collected an original dataset of 1,080 labelled samples across 36 classes (A–Z, 0–9) using a custom serial collection CLI, then expanded it 6× to 6,480 samples with a deterministic augmentation pipeline (jitter, magnitude scaling, time warping, axis rotation).
- Normalised variable-length IMU bursts into fixed 100×6 tensors and trained a stacked LSTM (128 → 64, dropout 0.3) in TensorFlow/Keras, reaching **100% accuracy on a 1,296-sample held-out test split** across all 36 classes.
- Wrote the ESP32 firmware in Arduino C++ — 50 Hz sampling, button-gated capture, JSON-over-serial framing, LED state indication and a 500-sample overflow guard.
- Shipped a Tkinter inference GUI with character → word → sentence building and offline text-to-speech, targeting hands-free and accessibility-oriented text entry.

Technology Stack : Python, TensorFlow/Keras, scikit-learn, NumPy, pandas, Arduino C++, ESP32-C3, MPU-6050, pySerial, Tkinter

FUNCTIONAL / TECHNICAL SKILLS

■ Programming Languages	: JavaScript, TypeScript, Python, Java, C++ (Arduino)
■ Frontend	: React 19, Tailwind CSS, shadcn/ui, Vite, TanStack Query, Capacitor (Android)
■ Backend	: Node.js, Express, REST API design, Socket.IO, WebRTC, JWT, webhooks, node-cron
■ Databases & Storage	: Firebase Firestore, PostgreSQL, MySQL, Backblaze B2 / AWS S3
■ AI / ML	: OpenAI API (vision, speech-to-text, TTS, tool calling), TensorFlow/Keras, scikit-learn
■ Integrations	: Cashfree Payments, WhatsApp Business API, Web Push (VAPID), Google Maps, Nodemailer
■ Cloud & Deployment	: Firebase (Auth, Admin SDK), Render, Vercel, Google Play (Capacitor builds)
■ Tools	: Git, GitHub, Postman, Appwrite, pdf-lib, Arduino IDE

ACHIEVEMENTS

- **14th place (Top 20)** — OpenAI Codex Hackathon, OpenAI × NamasteDev (2026)
- **99.11 percentile** — MHT MCA CET (2025)
- **Top 1%** — NPTEL Data Structures & Algorithms exam (85%, Elite + Silver)
- **2nd Place** — Kartikeya Rindhani Memorial Coding Competition (2025)